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Cambro-Ordovician palaeomagnetic and geochronologic data from southern Victoria Land, Antarctica: revision of the Gondwana apparent polar wander path

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SUMMARY

We present new palaeomagnetic and isotopic data from the southern Victoria Land region of the Transantarctic Mountains in East Antarctica that constrain the palaeogeographic position of this region during the Late Cambrian and Early Ordovician. A new pole has been determined from a dioritic intrusion at Killer Ridge ($^{40}Ar/^{39}Ar$ biotite age of 499 ± 3 Ma) and hornblende diorite dykes at Mt. Loke (21°E, 7°S, $A95 = 8°$, $N = 6$ VGPs). The new Killer Ridge/Mt. Loke pole is indistinguishable from Gondwana Late Cambrian and Early Ordovician poles. Previously reported palaeomagnetic poles from southern Victoria Land have new isotopic age constraints that place them in the Late Cambrian rather than the Early Ordovician. Based upon the new palaeomagnetic and isotopic data, new Gondwana Late Cambrian and Early Ordovician mean poles have been calculated.

Key words: Cambrian, East Antarctica, Gondwana, Ordovician, palaeomagnetism.

INTRODUCTION

The early Palaeozoic Gondwana apparent polar wander path has always had problematic segments, especially in the Cambrian. Part of the problem has been in knowing the age of the palaeomagnetic poles, and correlating poles dated by isotopic techniques with those dated biostratigraphically. In the last five years, substantial revisions have been made to the Cambrian timescale (Bowring *et al.* 1993; Tucker & McKerrow 1995; Bowring & Erwin 1998). Using the new Bowring & Erwin (1998) timescale, in which the Middle Cambrian ranges from 510 to ~500 Ma (see also Encarnación *et al.* 1999) and the Late Cambrian ranges from ~500 to ~490 Ma, we have revised the Gondwana Late Cambrian to Early Ordovician APWP based on new palaeomagnetic and isotopic data from southern Victoria Land. Using the revised poles, we hoped to further constrain the Gondwana apparent polar wander path (APWP).

REGIONAL GEOLOGY

The geology of southern Victoria Land (SVL) (Fig. 1) consists, in part, of a Proterozoic metamorphic basement of upper amphibolite facies schists and marbles (Adams & Whitla 1991) intruded by Vendian to Ordovician granitoids and dykes (Allibone *et al.* 1991; Rowell *et al.* 1993; Encarnación & Grunow 1996). The Cambro-Ordovician calc-alkaline granitoids are thought to be associated with westward-dipping subduction beneath Antarctica, while the youngest granitoid suite may reflect uplift and extension in SVL during the Middle Ordovician (Allibone *et al.* 1993). The metamorphic basement and the lower Palaeozoic intrusions are overlain unconformably by the essentially flat-lying Beacon Group sedimentary rocks (Devonian to Triassic) (McKelvey & Webb 1962). Ferrar Group diabase sills and feeder dykes (~183 Ma) intrude the previously mentioned rock types (Allibone *et al.* 1991; Encarnación *et al.* 1996). At localized areas along the mainland of southern Victoria Land there are small, late Cenozoic alkaline volcanic centres (the McMurdo Volcanic Group) (Allibone *et al.* 1991).

PALAEOMAGNETIC PROCEDURES

Previous palaeomagnetic and isotopic results from the southern Victoria Land area have been published in Grunow (1995) and Encarnación & Grunow (1996). Six to seven cores were usually sampled at each site using a Pomeroy gasoline-powered rock-coring drill. The samples were oriented with a Brunton magnetic compass. Sun compass measurements were made at each locality to allow correction for local magnetic variation and were always within 0° to 3° of the predicted magnetic variation. The geographical coordinates of some locations were determined with a handheld GPS 'Magellan' receiver and are listed to 0.01° in Table 1. Analyses of specimens were performed

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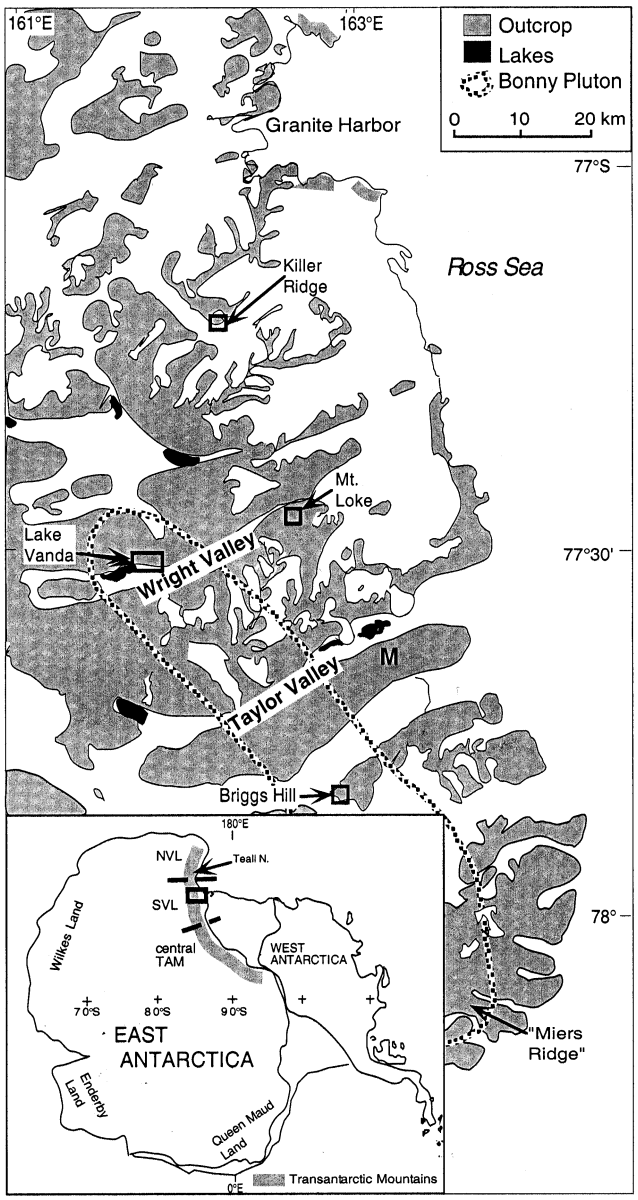


Figure 1. Palaeomagnetic locations in southern Victoria Land. M is the palaeomagnetic study area of Manzoni & Nanni (1977). Location of Bonny pluton from Cox (1993). Inset: locations of other palaeomagnetic studies of Early Palaeozoic rocks in East Antarctica; box represents location of southern Victoria Land.

at the Byrd Polar Research Center palaeomagnetism laboratory. A JR5A magnetometer was used to measure the natural remanent magnetization (NRM) on specimens from each core. Pilot samples were progressively demagnetized using stepwise alternating field (AF) or thermal (TH) techniques to determine the best demagnetization procedure for each site. During AF demagnetization, steps of generally 5–10 mT up to a peak of 90 mT were used (minimum of eight steps), while during thermal demagnetization steps of generally 50 °C or 100 °C up to 500 °C and then smaller intervals were used (minimum of 10 steps). Susceptibility measurements were made on a Bartington MS2 susceptiblity bridge after each thermal demagnetization step to evaluate mineralogical changes associated with heating the specimens. Isothermal remanent magnetization (IRM) experiments were performed on representative specimens from the main sample locations.

The magnetization directions were determined by principal component analysis for collinear segments converging towards the origin (Kirschvink 1980) plotted on vector endpoint diagrams. Magnetization directions were considered stable (whether primary or secondary) if defined by three or more collinear points on the vector component diagrams and produced maximum angular deviation values for linear segments of less than 15° (usually less than 10°). The mean characteristic magnetization directions for each locality were obtained by combining sample means to make site means and then site means to make a unit mean using standard Fisher (1953) statistics. Site mean directions with $\alpha 95 \geq 15°$ were not used for calculating unit mean directions or virtual geomagnetic poles (VGPs).

RESULTS

Killer Ridge

The Killer Ridge hornblende–biotite diorite is exposed along ~400 m of outcrop (Fig. 1) (Gunn & Warren 1962). No contacts were observed with the other intrusive rocks in the area. Two sites were drilled in the diorite. Both sites yielded stable magnetic directions that were very shallowly upward pointing to the south-southwest (Table 1; Fig. 2a). A few samples from both sites were of opposite polarity and these samples contained a haematitic component that was not removed until 630 °C. We believe that this higher temperature component is probably due to deuteric alteration of the intrusion near to the time of emplacement, although later low-grade regional metamorphism cannot be ruled out. In most samples, the remanent magnetization was removed by 580 °C.

A $^{40}Ar/^{39}Ar$ biotite plateau age of 499 ± 3 Ma (2σ) (MSWD = 0.3) was obtained on samples from the first palaeomagnetic site (KIL3A) (Table 2; Fig. 3). The plateau consists of the three highest T steps (Table 2) and has nearly a third of the ^{39}Ar released. Steps 4–11 have a 'plateau' age of 503 ± 4 Ma, which is indistinguishable from the age of the three high-T steps, but shows greater scatter (MSWD = 11). The hornblende in this rock contained finely intergrown biotite making the hornblende unsuitable for $^{40}Ar/^{39}Ar$ dating.

Mt. Loke

Mt. Loke exposes numerous mafic dykes intruding a deformed granitic pluton. The oldest and most prevalent of the dyke swarms is composed of hornblende–biotite diorite dykes and bosses (Ellery 1989). The slightly altered diorite dykes are equigranular, medium-grained dykes of up to 20 m width with no significant chilled margins (Ellery 1989). Cutting the E–W-trending diorite dykes are NE-trending mafic and granite dykes (Keiller 1988). The diorite dykes are thought to be related to an earlier phase of dioritic intrusion in the area and to the emplacement of the Denton pluton (Ellery 1989).

Four diorite dyke sites were drilled at Mt. Loke and all yielded stable magnetic directions that were shallowly upward pointing to the south-southwest (Table 1). The NRM was completely removed by 580 °C, in agreement with IRM experiments indicating that the probable magnetic carrier is a titanomagnetite.

Table 1. New southern Victoria Land site mean directions and virtual geomagnetic poles.

Site coordinates Long.°E,	Lat.°S	Site	Isotropic age	Lithology	NRM × 10^{-2} A/m	Samples *n*/*N*	*D*(°)	*I*(°)	*a*95	*k*	VGP Long.°E	Lat.°S
Killer Ridge												
162.19,	77.21	KIL3A	~499 ± 3	diorite	3.6	6/6	212.1	−2.9	11.5	35	14.8	−9.4
162.19,	77.21	KIL3B	~499 ± 3	diorite	3.5	7/7	222.0	−4.2	7.4	67	24.6	−7.4
Mt. Loke												
162.51,	77.49	LK1A	>484 ± 7, ~499 ± 3?	diorite	0.12	7/7	215.1	−3.3	6.8	79	18.0	−8.6
162.51,	77.49	LK1B	>484 ± 7	mafic dyke	2.57	4/6	221.6	−4.4	6.5	201	24.5	−7.2
162.51,	77.49	LK1C	>484 ± 7, ~499 ± 3?	diorite	0.24	7/7	229.8	−28.3	3.8	251	30.4	−6.9
162.51,	77.49	LK1G	>484 ± 7, ~499 ± 3?	diorite	0.40	6/7	218.0	12.5	3.2	450	22.1	−16.1
162.51,	77.49	LK1H	>484 ± 7, ~499 ± 3?	diorite	0.29	7/7	215.9	−5.8	5.2	135	18.7	−7.2
All Diorites			~499 ± 3?	diorite		6	218.5	−5.5	12.0	32.0	21.4	−7.0 *A*95 = 7.7°

Site locations are shown in Fig. 1. *n*/*N*: number of samples used for site mean calculation/total number of samples; *D*: *in situ* site mean declination; mean *I*: *in situ* site inclination; *a*95: semi-angle of the cone of 95 per cent confidence about the site mean; *k*: estimate of precision parameter; VGP: virtual geomagnetic pole; positive values are in the Northern Hemisphere, negative values in the Southern Hemisphere. Present-day field direction for the sample area is declination ~152°E, inclination −82°. Isotopic age data are discussed in the text.

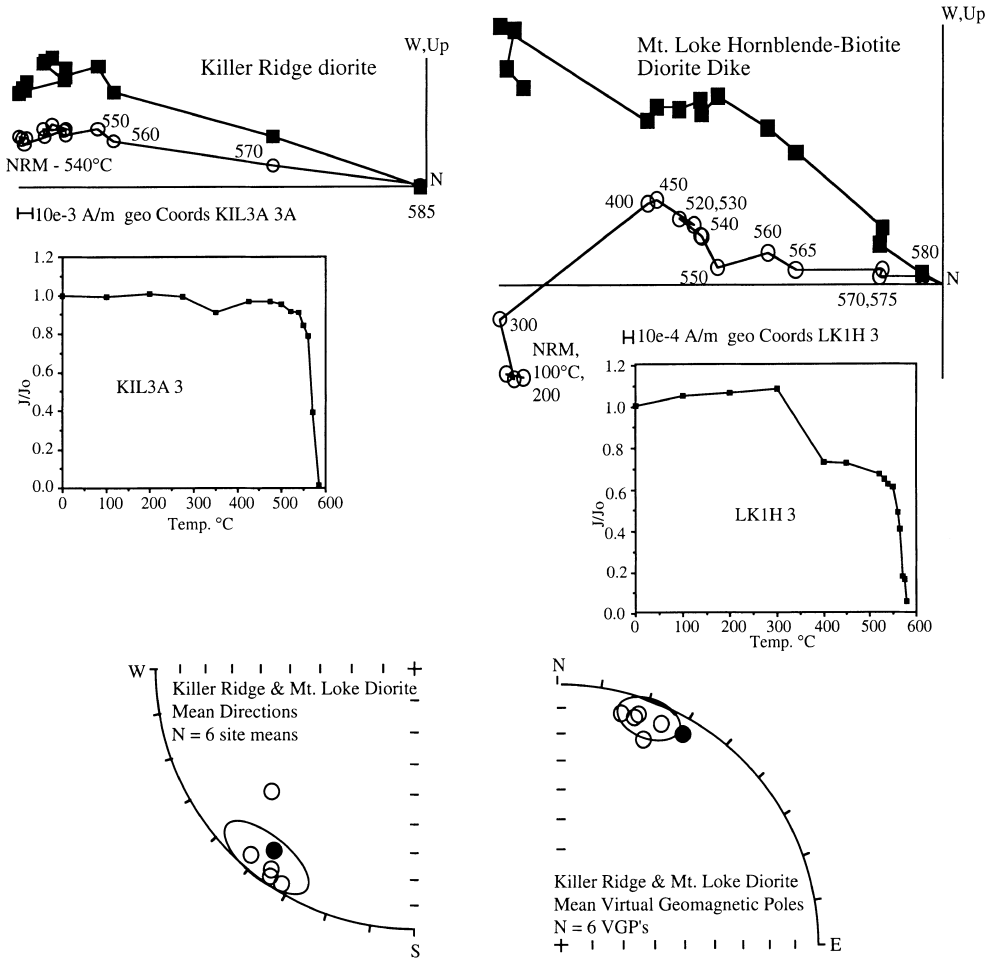


Figure 2. Orthogonal projection of vector endpoint diagrams showing demagnetization behaviour of samples from SVL. Open circles (solid circles) are projections on the vertical (horizontal) planes at the indicated levels of AF (mT) or thermal (°C) demagnetization. Normalized intensity (J/J_0) plots are shown for lithologies where thermal demagnetization was used. Equal-angle projections of site mean directions and VGPs from the Killer Ridge and Mt. Loke diorites. Open (closed) symbols are on the upper (lower) hemisphere. Circle represents the *a*95 (*A*95) radius of 95 per cent confidence for the directional (VGP) mean.

Table 2. $^{40}Ar/^{39}Ar$ results from southern Victoria Land.

Step	T (°C)	$\%^{39}Ar$	$\%^{40}Ar^*$	Age (Ma)	1σ (Ma)
KIL-3A (biotite)					
1	500	1.74	52.4	109.9	0.98
2	650	4.97	91.8	452.8	1.35
3	700	7.93	98.8	491.8	1.36
4	750	7.86	99.2	499.7	1.47
5	800	9.58	99.3	502.6	1.39
6	850	8.64	99.3	506.1	1.44
7	900	10.10	99.3	510.7	1.40
8	975	17.24	99.5	504.9	1.45
9	1075	23.19	99.7	499.2	1.38
10	1200	7.38	99.7	497.9	1.37
11	1400	1.38	99.6	497.9	1.37
Plateau (steps 9–11) = 499 ± 3 Ma, MSWD = 0.3					
GH-5C (hornblende)					
1	700	4.84	47.5	300.4	4.81
2	800	3.37	87.5	505.0	3.44
3	875	3.03	94.3	516.0	1.66
4	950	5.01	93.7	534.6	1.80
5	1025	34.29	94.9	508.8	1.68
6	1075	11.44	96.8	500.7	1.73
7	1125	14.93	97.7	497.6	1.58
8	1200	10.95	97.8	498.0	2.10
9	1500	12.13	92.6	497.2	1.97
Plateau (steps 6–9) = 498 ± 3 Ma, MSWD = 0.82					
VN-1L (hornblende)					
1	700	1.41	58.0	490.1	8.54
2	850	2.45	93.6	434.9	3.44
3	950	3.48	96.0	469.2	1.61
4	1000	15.06	98.5	497.9	1.45
5	1025	16.88	98.5	498.0	1.43
6	1050	11.73	98.9	494.8	1.41
7	1075	4.82	99.2	496.5	2.30
8	1120	12.62	98.6	500.2	1.45
9	1250	30.40	99.2	501.4	1.39
10	1500	1.14	93.8	493.3	4.02
Plateau (steps 4–9) = 499 ± 3 Ma, MSWD = 2.7					

$\%^{39}Ar$ is the amount of ^{39}Ar in each step out of the total ^{39}Ar in the sample. $\%^{40}Ar^*$ is the percent radiogenic ^{40}Ar of the total ^{40}Ar in the given step.

NEW ISOTOPIC DATA

We undertook $^{40}Ar/^{39}Ar$ dating to constrain the age of magnetization of several lithologies sampled for palaeomagnetism in southern Victoria Land. Many of the rocks did not have any isotopic constraints and some lithologies had U–Pb zircon or Rb/Sr ages (Encarnación & Grunow 1996). The $^{40}Ar/^{39}Ar$ dating technique provided a way to estimate the time of magnetization by comparing the closure temperature of various dated minerals to the unblocking temperature of the magnetization.

Apart from the new $^{40}Ar/^{39}Ar$ biotite age of 499 ± 3 Ma from Killer Ridge, we have obtained new $^{40}Ar/^{39}Ar$ ages from the Granite Harbor and Lake Vanda areas (Figs 1 and 3). The $^{40}Ar/^{39}Ar$ ages reported here have been corrected with an internal standard with an assigned age equivalent to 523.5 Ma for the standard MMhb-1 (Renne *et al.* 1994). The previously undated mafic dykes at Granite Harbor yielded an $^{40}Ar/^{39}Ar$ hornblende plateau age of 498 ± 3 Ma (2σ) (~50 per cent of the ^{39}Ar released, including the highest temperature steps; MSWD = 0.8) from palaeomagnetic site GH5C of Grunow (1995). The mafic dykes appear to be comagmatic with the grey Granite Harbor granite based on field relationships (Grunow 1995; Encarnación

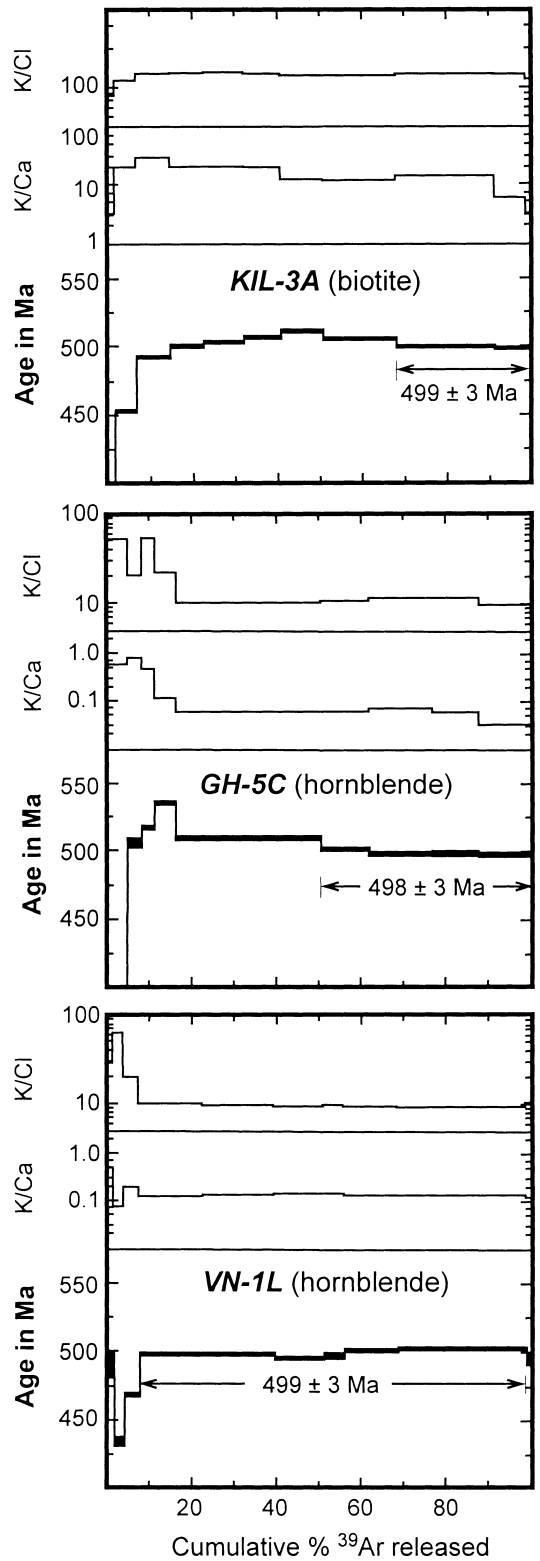


Figure 3. $^{40}Ar/^{39}Ar$ age spectra plots of mineral separates from Killer Ridge (KIL-3A), Granite Harbor mafic dyke (GH-5C) and the Bonny Pluton at Lake Vanda (VN-1 L). See Fig. 1 for locations.

& Grunow 1996). U–Pb zircon dating on the Granite Harbor pink granite yielded an age of 498 ± 4 Ma (Encarnación & Grunow 1996). These ages suggest that the Granite Harbor granites and dykes are approximately the same age and were emplaced in the Late Cambrian (or latest Middle Cambrian).

U–Pb zircon dating on the Bonny Pluton (Fig. 1) yielded an age of 505 ± 2 Ma from Miers Ridge (Fig. 1) (Encarnación & Grunow 1996). A new $^{40}Ar/^{39}Ar$ hornblende plateau age of 499 ± 3 Ma ($\sim$90 per cent of the highest-temperature ^{39}Ar released; MSWD = 2.7) was obtained from the Bonny Pluton at Lake Vanda (Fig. 3).

Correlation of southern Victoria Land dykes

The palaeomagnetic samples collected from microlamprophyric mafic dykes in the Taylor and Wright valleys (Manzoni & Nanni 1977; Grunow 1995) (Fig. 1) are geochemically similar (Vennum 1990). This mafic dyke swarm, called the dry valley mafic dyke swarm (DVMD) (Vennum 1990), is considered to be comagmatic with the Vanda feldspar porphyry dykes at Lake Vanda (Allibone *et al.* 1993), which have a U–Pb zircon age of 484 ± 7 Ma (Encarnación & Grunow 1996). However, the Granite Harbor mafic dykes do not appear to be similar to the DVMD swarm and they have older isotopic ages (498 ± 3 Ma $^{40}Ar/^{39}Ar$ hornblende plateau age).

While there are no direct isotopic age constraints on the Mt. Loke diorite dykes they are geochemically dissimilar to the DVMD described by Vennum (1990). The Mt. Loke diorite dykes also have a different, but consistent, structural trend of E–W instead of the relatively constant NE trend of the DVMD swarm, indicative of a different stress regime at the time of their emplacement. A dioritic plug at Lake Vanda in the Wright Valley that is similar to the Mt. Loke diorite dykes is truncated by the Vanda feldspar porphyry and mafic dykes (Ellery 1989). All of these observations can be interpreted to suggest that the Mt. Loke diorite dykes are older than 484 ± 7 Ma. Several workers believe that the small gabbroic and dioritic intrusions found in the region, including the diorite dykes and intrusion at Mt. Loke and the Killer Ridge diorite intrusion (499 ± 3 Ma), are probably related to each other (Gunn & Warren 1962; Allibone *et al.* 1993), hence we suggest that the Mt. Loke dykes were emplaced at $\sim$499 Ma.

REVISION OF SOUTHERN VICTORIA LAND PALAEOPOLES

Given the new $^{40}Ar/^{39}Ar$ ages presented in this paper and the previously reported U–Pb zircon ages (Encarnación & Grunow 1996), the age of the SVL mean poles published by Grunow (1995) have been revised. Shown in Table 3 are the SVL mean directions, palaeopoles and new age determinations. We have revised the palaeopole listed for the Bonny Pluton in Grunow (1995) based on the new age determinations. The Bonny Pluton palaeomagnetic sites from Lake Vanda have yielded an $^{40}Ar/^{39}Ar$ hornblende plateau age of 499 ± 3 Ma, making the magnetization age of these rocks probably Late Cambrian. The Bonny pluton at Briggs Hill yielded different directions from the Lake Vanda sites (Fig. 1, Table 3). Perhaps the Briggs Hill sites reflect a magnetization age closer to the 505 ± 2 Ma U–Pb zircon age obtained from Miers Ridge (Encarnación & Grunow 1996) rather than the 499 ± 3 Ma age from Lake Vanda. Alternatively, there may have been structural disruption of the Briggs Hill site but this has not been observed in the field mapping.

Based on the similarity in lithology, outcrop description and field relationships, we believe that the Mt. Loke dioritic dykes and intrusions are probably part of the same intrusive event that emplaced the Killer Ridge dioritic plug. Hence, the

Table 3. Southern Victoria Land Cambrian and Ordovician palaeomagnetic poles.

Formation/Unit	Age†		*n*/*N*	Dec	Inc.	*a*95	PPLong	PPLat	*A*95	*dp*, *dm*	*Q*	Reference
Lake Vanda	Fdsp.porphyry dykes	484 ± 7	38/7	225.4°	−9.6°	8.5°	27.2°	−3.9°	7.2°		5	Grunow (1995)
Lake Vanda	Bonny Pluton	499 ± 3	28/4	225.5°	−0.9°	13.1°	27.9°	−8.3°	11.9°		4	Grunow (1995)
Taylor Valley	Lamprophyre Dykes	$\sim 484 \pm 7$	60/4	222.6°	0.6°	10.9°	26.7°	−9.3°	11.0°		3	Manzoni & Nanni 1977
Granite Harbor	Pink Granite	498 ± 4	21/3	214.1°	−14.9°	8.3°	16.7°	−3.4°	–	4.3°, 8.5°	3	Grunow (1995)
Granite Harbor	Mafic Dykes	498 ± 3	27/5	213.1°	−19.5°	8.6°	15.0°	0.8°	7.5°		4	Grunow (1995)
Granite Harbor	Grey Granite	$\sim 498 \pm 3$	34/5	219.3°	−12.8°	6.4°	21.6°	−3.6°	4.0°		4	Grunow (1995)
Briggs Hill	Bonny Pluton	505 ± 2	8/1	239.5°	−17.6°	14.5°	41.4°	2.7°	–	7.8°, 15°	3	Grunow (1995)
Mt. Loke	Hornblende Diorite	$> 484 \pm 7$; $\sim 499 \pm 3$?	27/4	219.4°	−6.2°	20.8°	22.3°	−6.3°	12.7°		3	this paper
Killer Ridge	Diorite	499 ± 3	13/2	217.5°	−3.6°	6.4°	20.1°	−8.3°	–	3.2°, 6.4°	3	this paper
Killer Ridge/ Mt. Loke	Diorites	$\sim 499 \pm 3$	40/6	218.5°	−5.5°	12.0°	21.4°	−7.0°	−7.7°			this paper

n/*N*: total number of samples represented/number of sites used for calculation of mean direction and palaeopole; underscore: number of sites (samples) used to calculate listed means; *D*: declination; *I*: inclination; *a*95: error circle at the 95 per cent confidence level; k: estimate of precision parameter; PPLon and PPLat: palaeopole longitude and latitude; A95: radius of error circle at 95 per cent confidence level; *dp*, *dm*: confidence ellipse about the pole; *Q*: quality factor based on reliability criteria described in Van der Voo (1988).
† Isotopic ages are either discussed in the text or are found in Encarnacion & Grunow (1996).

palaeomagnetic results from Killer Ridge and Mt. Loke have been considered together. The Late Cambrian Killer Ridge/Mt. Loke pole is very similar to the Late Cambrian and Early Ordovician data from southern Victoria Land (Table 3; Fig. 4).

Although structural disruption or tilting of the SVL intrusive rocks listed in Table 2 may be possible, we doubt that this has significantly affected our data. The intrusions are from several different fault blocks over a large area (Wilson 1992). Significant tilting has not occurred in southern Victoria Land since the Devonian because flat-lying Beacon strata rest non-conformably on the intrusive and metamorphic basement. Tilting may have occurred prior to the Devonian but the consistency of the data over such a large area and within temporally similar but lithologically different intrusions sampled across known fault blocks argues against that. Lanza & Tonarini (1998) suggested that there is a regional westward tilt of 2°–5° over the SVL and northern Victoria Land regions. We have not adjusted the palaeomagnetic poles for this suggested tilting, because it would make only a very small difference in the pole analysis. In addition, the slight dip of the Beacon strata may be partly the result of the thick Ferrar sills found intruding the Beacon throughout the area but not observed to intrude the basement rocks except near the contact with the Beacon. All of the SVL samples listed in Table 3 come from below any known Ferrar sills.

COMPARISON WITH GONDWANA POLES

In Table 4, a list of Gondwana Middle Cambrian to Early Ordovician poles is shown (rotated into an African reference frame). We have revised the age assignment of the Lake Vanda Bonny Pluton and Granite Harbor poles from Grunow (1995) to be Late Cambrian rather than Early Ordovician. A few of the Cambrian to Early Ordovician Antarctic poles listed by Lanza & Tonarini (1998) have been omitted from Table 4 due to the uncertainty in the age of the magnetization, a Q factor of less than 3 and/or insufficient information about the demagnetization procedures.

There are no significant differences between the Gondwana Late Cambrian poles (~500–490 Ma) and those from the Early Ordovician (~485 Ma) (Table 4), apart from the Australian Chatswood limestone and Black Hill norite poles (Ripperdan & Kirschvink 1992; Schmidt *et al.* 1993). It is unclear why these two poles are so different from the other Gondwana Late Cambrian and Early Ordovician poles. The Chatswood and Black Hill poles are similar to the Middle Ordovician and Silurian Gondwana poles, respectively (Grunow 1999), but there is no evidence that these rocks have been remagnetized.

Eliminating the Chatswood and Black Hill norite poles results in Gondwana Late Cambrian and Early Ordovician mean poles from more than one continent that are in good

Table 4. Gondwana Middle Cambrian to Middle Ordovician palaeomagnetic poles (in an African reference frame).

Formation/Unit		Age (Ma)†	PPLong	PPLat	*A*95	*Q*	Reference
Early Ordovician (490–478)							
Australia	Black Hills High Temp.*	487 ± 5	12.0°	13.0°	3.0°	4	Schmidt *et al.* (1993)
Australia	Black Hills High Coercivity*	487 ± 5	357.0°	13.0°	7.0°	4	Schmidt *et al.* (1993)
Australia	Jinduckin	Lower Ord.	0.0°	36.0°	13.0°	4	Luck (1972)
Africa	Graafwater Fm.	Lower Ord.	14.0°	28.0°	9.0°	6	Bachtadse *et al.* (1987)
Antarctica	Felsite Dykes	484 ± 7	5.0°	42.0°	6.0°	5	Grunow (1995)
Antarctica	Taylor V. Mafic Dykes	$\sim 484 \pm 7$	8.0°	36.0°	11.0°	4	Manzoni & Nanni (1977)
Antarctica	Teall Nunatak	471–487	344.0°	46.0°	7.0°	5	Lanza & Tonarini (1998)
		MEAN (~485 Ma), $N = 5$	3.0°	38.0°	11.0°		this paper
		MEAN (~485 Ma), $N = 4$	7.0°	36.0°	9.0°		this paper without Teall Nunatak
Late Cambrian (~500–490 Ma)							
Australia	Chatswood Lmst*	Upper Camb.	41.0°	51.0°	10.0°	5	Ripperdan & Kirschvink (1992)
Australia	Dundas Gp	Upper Camb.	350.0°	23.0°	12.0°	3	Giddings & Embleton (1974)
Australia	Tasmania	Upper Camb.	5.0°	31.0°	5.0°	6	Li *et al.* (1997)
Antarctica	Granite H. Pink Granite	$<498 \pm 3$	355.0°	33.0°	–	4	Grunow (1995)
Antarctica	Granite H. Lamp. Dykes	498 ± 3	354.0°	35.0°	8.0°	5	Grunow (1995) and this paper
Antarctica	Granite H. Grey Gr.	$>498 \pm 3$	359.0°	37.0°	4.0°	5	Grunow (1995)
Antarctica	Bonny Pluton, Lake Vanda	499 ± 3	9.0°	37.0°	12.0°	4	Grunow (1995)
Antarctica	Killer Ridge/Mt.Lok	499 ± 3	2.0°	34.0°	8.0°	4	this paper
		MEAN (~495 Ma), $N = 7$	0.0°	34.0°	6.0°		this paper
Middle Cambrian (~510–500 Ma)							
Australia	Basal Lake Frome	Middle Camb.	4.0°	20.0°	13.0°	5	Klootwijk (1980)
Australia	Areyonga	Middle Camb.	352.0°	13.0°	5.0°	6	Klootwijk (1980)
Australia	Hudson	Middle Camb.	332.0°	63.0°	13.0°	3	Luck (1972)
Australia	Hugh River	Middle Camb.	13.0°	62.0°	9.0°	3	Embleton (1972); Lindsay (1987)
Antarctica	Bonny Pluton, Briggs Hill	505 ± 2?	16.0°	54.0°	–	3	Grunow (1995) and this paper

PPlong, PPlat: palaeopole longitude and latitude, respectively. Only poles with Q ratings of 3 or higher are listed (see Van der Voo 1988 for criteria). Q ratings for poles published prior to 1992 are from Van der Voo (1993).

* Poles not used to calculate the Gondwana mean poles.

† Numeric ages for the poles are based on the Bowring & Erwin (1998) timescale.

The following are the rotation poles used to bring all poles into an African reference frame:

Royer & Sandwell (1989): Australia to Antarctica 38.9°E, −2.0°S, −31.5°

Lawver & Scotese (1987): Antarctica to Africa 328.58°E, −7.78°S, 58.0°.

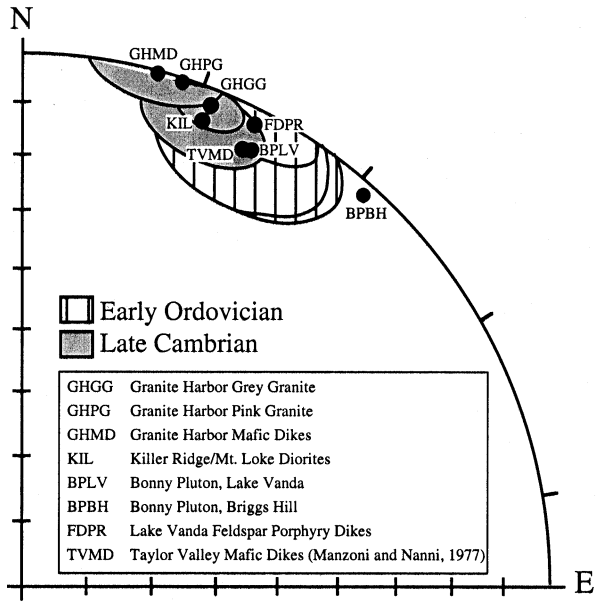


Figure 4. Equal-angle projections of VGPs from southern Victoria Land (Table 3). Circles represent the *A*95 radius of 95 per cent confidence for the various location means.

agreement (Table 4). This suggests relatively little apparent polar wander during this 20 Myr time interval, or a pervasive, synchronous remagnetization event in Antarctica and Australia (Fig. 5). We do not think the Late Cambrian poles have been remagnetized in the Ordovician because the Australian poles contain dual polarities and/or positive fold tests. While the Antarctic poles do not have fold tests (because they are intrusions), the poles come from very different lithologies (diorites, felsic dykes, mafic dykes and several different granites) that exhibit only minor alteration. The $^{40}Ar/^{39}Ar$ age dating indicates that there has been no Ordovician thermal overprinting event of the Late Cambrian igneous rocks to cause remagnetization.

The Antarctic Teall Nunatak pole (Lanza & Tonarini 1998) is a little different from the other Gondwana Early Ordovician poles (Table 4). Perhaps this is because these rocks may be Middle Ordovician rather than Early Ordovician in age, given the Rb–Sr whole rock–biotite ages of 475 ± 4 and 483 ± 4 Ma (Lanza & Tonarini 1998). The Teall Nunatak pole is indistinguishable from a pole for Middle to Late Ordovician sediments in Australia (343°, 52°N, $A95 = 5°$ in an African reference frame) (Goleby 1980).

We have tentatively listed the Briggs Hill Bonny Pluton pole as Middle Cambrian (based on the 505 Ma U–Pb zircon

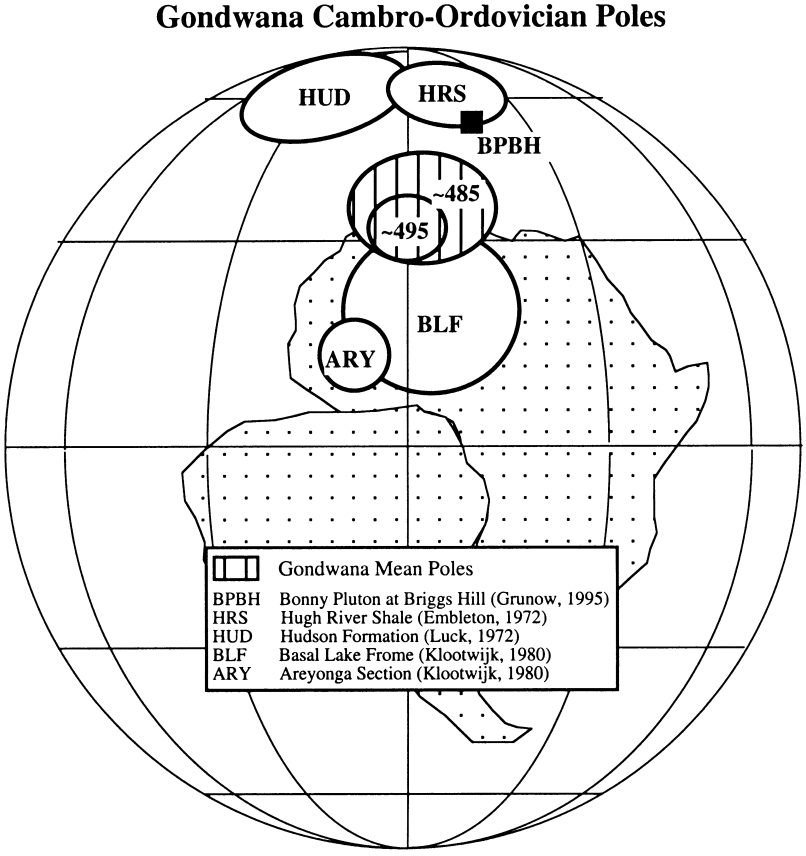


Figure 5. Gondwana Late Cambrian and Early Ordovician mean poles (from Table 4) compared with individual Middle Cambrian poles from Australia and Antarctica. Note the similarity between the Late Cambrian and Early Ordovician mean poles but the disparity between various Middle Cambrian poles. While there is no evidence for a Late Cambrian to Early Ordovician loop in the Gondwana apparent polar wander path, there could be a loop if the higher-latitude Middle Cambrian Gondwana poles are correct. Circles represent the *A*95 radii of 95 per cent confidence for the poles.

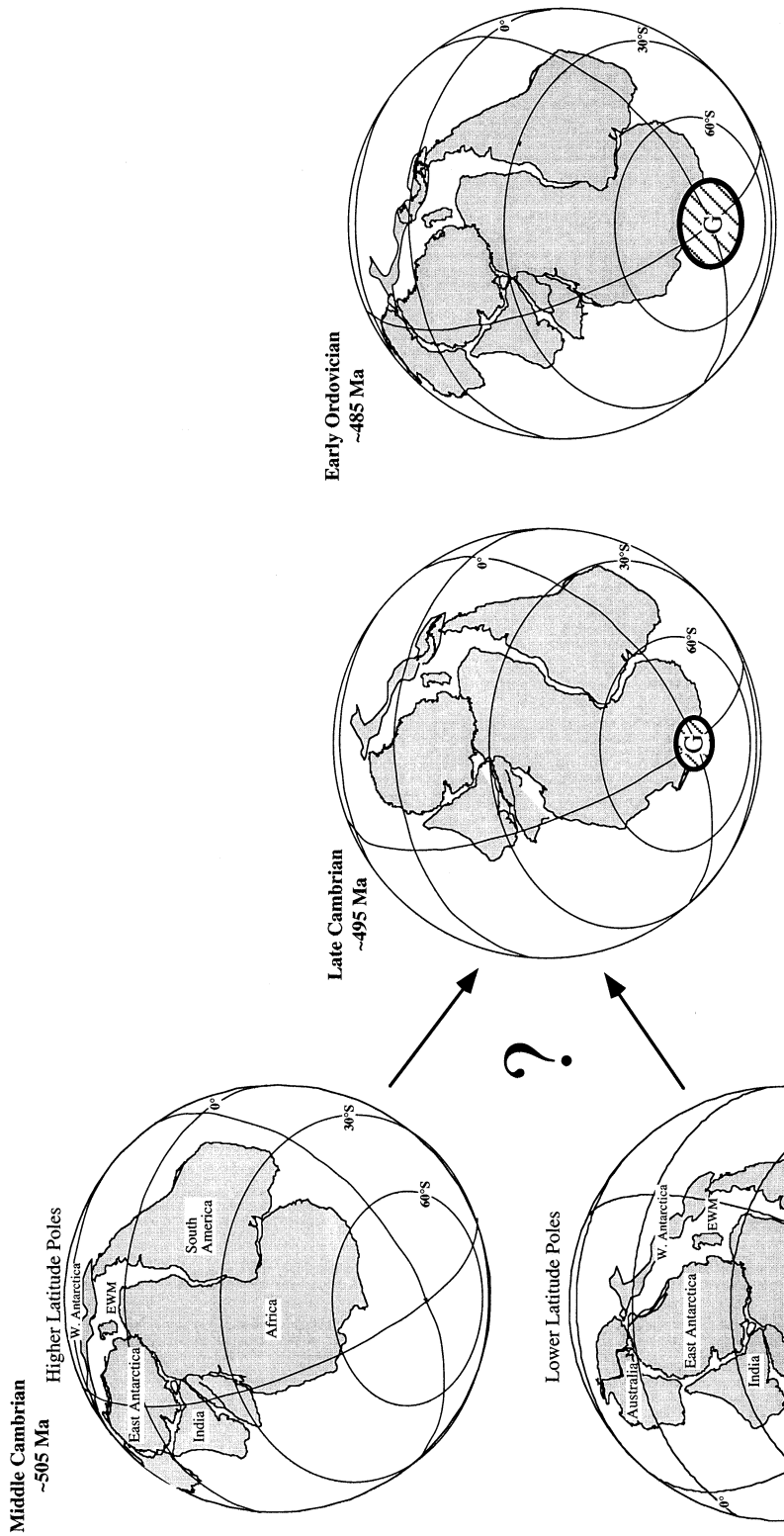


Figure 6. Plate reconstructions of Gondwana for the Late Cambrian and Early Ordovician using the mean poles listed in Table 4. The geographical south pole is centred on the Gondwana reference pole with its A_{95} circle. The Middle Cambrian reconstructions show the approximate position of Gondwana using higher- and lower-latitude Middle Cambrian poles. There is no $A95$ for the Middle Cambrian poles because there are not enough poles in agreement to define a mean. Note that there is no palaeomagnetically discernible difference in position for Gondwana between the Late Cambrian and Early Ordovician. The continental fragments of West Antarctica and New Zealand have tentatively been shown in their Mesozoic pre-Gondwana break-up positions relative to East Antarctica.

age) given that it is similar to some other Gondwana Middle Cambrian poles. However, none of the Middle Cambrian Gondwana poles listed in Table 4 is of particularly high quality using the quality (Q) criteria developed by Van der Voo (1988). The Middle Cambrian poles appear to lie in two groups (Fig. 5). Two Australian sedimentary sequences yield results that place the poles in central west Africa, while two other poles from Australia (Luck 1972; Embleton 1972) are located north of Africa (Table 4; Fig. 5). The Bonny Pluton pole from Briggs Hill presented here (Tables 3 and 4) may lend some support to higher-latitude Australian poles.

GONDWANA PLATE RECONSTRUCTIONS

The revision of the Cambrian timescale (Bowring & Erwin 1998) has caused changes in the Gondwana Cambrian polar wander path and consequently previous plate reconstructions also need to be updated (Scotese & McKerrow 1990; Grunow 1995). The Late Cambrian and Early Ordovician reconstructions are virtually identical and reflect little apparent polar wander during this interval (Fig. 6). The uncertainty about the Middle Cambrian mean pole position for Gondwana indicates that many more new poles are needed to constrain any Middle Cambrian plate reconstructions.

CONCLUSIONS

The new palaeomagnetic and isotopic data presented here, along with the revised Cambrian timescale (Bowring & Erwin 1998), has resulted in new Gondwana Late Cambrian and Ordovician mean poles. The Late Cambrian and Early Ordovician mean poles are indistinguishable from each other, which may indicate either little apparent polar wander or a regional, synchronous remagnetizaton event, although the latter seems unlikely. The Gondwana Middle Cambrian poles remain problematic and require additional data to test whether there was significant polar wander between the Cambrian and Early Ordovician periods.

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